

Workshop Title

OpenBIM and Intelligent Building Modelling

Workshop Description

This workshop introduces participants to OpenBIM as a structured way of understanding buildings, rather than as a software-driven activity, and focuses on developing conceptual clarity for professionals and students who are new to BIM.

Participants will understand why BIM exists, how buildings are modelled digitally, how information is structured and exchanged, and how building data can be extracted and evaluated for multiple use cases using code. The sessions move from fundamentals to hands-on modelling and data-driven evaluation, using open standards and neutral workflows.

By the end of the workshop, participants will be able to relate digital building models to real-world project needs, understand OpenBIM deliverables, and appreciate how structured information supports analysis, coordination, and informed decision-making.

Workshop Sessions

1. OpenBIM and Use Cases (1 hour)

Introduction to OpenBIM principles, the need for open standards, and real-world use cases across design, construction, and asset lifecycle stages, with emphasis on collaboration and information exchange.

2. Use of AI for Concept Design (0.5 hour)

Introduction to the role of artificial intelligence in early-stage design, demonstrating how AI can support concept generation, and decision-making during the initial phases of a project.

3. Integrated Building Modelling (2.5 hours)

Hands-on development of a simple 3D model, covering architectural elements, structural components, and basic building services, focusing on spatial logic, and coordination.

4. Library Creation (0.5 hour)

Creation of reusable building element libraries with structured information, highlighting consistency, reusability, and the role of libraries in efficient OpenBIM workflows.

5. OpenBIM Standard Deliverables (0.5 hour)

Understanding common OpenBIM deliverables, the type of information they contain, how they are structured, and how they support collaboration and downstream uses.

6. Data Extraction and Evaluation Using Code (1 hour)

Introduction to extracting and evaluating structured building data using code, demonstrating how OpenBIM information supports multiple project use cases..

Overall Learning Outcome

Participants develop the ability to interpret digital building models as structured information, understand OpenBIM-based workflows, and recognise how building data supports analysis, coordination, and informed decision-making.